COMPLETION OF A FINITE TYPE MORPHISM IS FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-map-completions-finite-type; the proof environment immediately following it is its own proof; the context span delivers the section situation and the cartesian square that the statement's phrase 'in the situation above' refers to

Editorial note. The label lemma-map-completions-finite-type names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the /T completion functor and the cartesian square relating f_{/T'} to f (section-completion-functor, delivered as context)
- base change preserves local finite type (spaces-morphisms-lemma-base-change-finite-type of Morphisms of Spaces)

Source reference. Original lemma 0APV: sources/stacks-chow/tags.txt, source lines 13488–13488. Prerequisite pointer invoked at 6069; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0AM3: sources/stacks-chow/tags.txt, source lines 13391–13391. Prerequisite pointer invoked at 6016; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03XH: sources/stacks-chow/tags.txt, source lines 5148–5148. Prerequisite pointer invoked at 6022; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

1. The completion functor

In this section we consider the following situation. First we fix a base scheme S. All rings, topological rings, schemes, algebraic spaces, and formal algebraic spaces and morphisms between these will be over S. Next, we fix an algebraic space X and a closed subset $T \subset |X|$. We denote $U \subset X$ be the open subspace with $|U| = |X| \setminus T$. Picture

$$U \to X \quad |X| = |U| \amalg T$$

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In this situation, given an algebraic space X' over X, i.e., an algebraic space X' endowed with a morphism $f : X' \to X$, then we denote $T' \subset |X'|$ the inverse image of T and we let $U' \subset X'$ be the open subspace with $|U'| = |X'| \setminus T'$. Picture

$$U' = f^{-1}U \qquad \begin{array}{ccc} U' & \longrightarrow & X' \\ \downarrow & & \downarrow{\scriptstyle f} \\ U & \longrightarrow & X \end{array} \qquad \begin{array}{ccccc} |U'| & \longrightarrow & |X'| & \longleftarrow & T' \\ \downarrow & & \downarrow{\scriptstyle |f|} & & \downarrow \\ |U| & \longrightarrow & |X| & \longleftarrow & T \end{array} \qquad T' = |f|^{-1}T$$

We will relate properties of f to properties of the induced morphism

$$f_{/T} : X'_{/T'} \longrightarrow X_{/T}$$

of formal completions. As indicated in the displayed formula, we will denote this morphism $f_{/T}$. We have already seen that $f_{/T}$ is representable by algebraic spaces in Formal Spaces, Lemma 0APV. In fact, as the proof of that lemma shows, the diagram

$$\begin{array}{ccc} X'_{/T'} & \longrightarrow & X' \\ {\scriptstyle f_{/T}}\downarrow & & \downarrow{\scriptstyle f} \\ X_{/T} & \longrightarrow & X \end{array}$$

is cartesian. Please keep this fact in mind whilst reading the lemmas stated and proved below.

Lemma 1.1. *In the situation above. If f is locally of finite type, then $f_{/T}$ is locally of finite type.*

Proof. (Finite type morphisms of formal algebraic spaces are discussed in Formal Spaces, Section 0AM3.) Namely, suppose that $Z \to X$ is a morphism from a scheme into X such that $|Z|$ maps into T. From the cartesian square above we see that $Z \times_X X'$ is an algebraic space representing $Z \times_{X_{/T}} X'_{/T'}$. Since $Z \times_X X' \to Z$ is locally of finite type by Morphisms of Spaces, Lemma 03XH we conclude. □